CIE AS PHYSICS (9702) • MARK SCHEME & MODEL ANSWERS

3.3 Linear Momentum & its Conservation

Syllabus Section: 3. Dynamics **Paper Type:** Multiple Choice (Paper 1) **Total:** 34 Questions (34 Marks)

Question 1 Easy [1 mark]

What is the principle of conservation of momentum?

A force is equal to the rate of change of momentum

B momentum is the product of mass and velocity

C the total momentum of two bodies after collision is equal to their total momentum before collision

D the total momentum of a system remains constant provided no external force acts on it

Mark Scheme & Explanation:

*The correct answer is **D** because:*

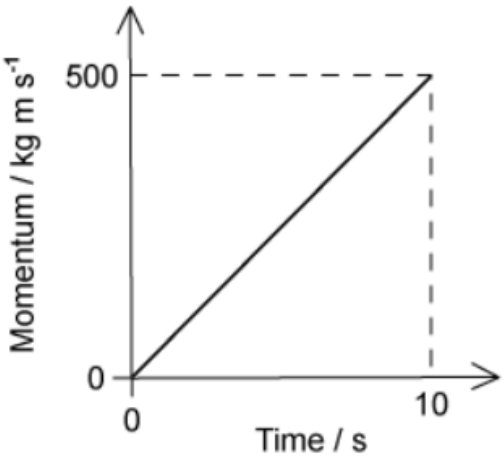
- The principle of conservation of momentum states that:
  - For a system of interacting bodies, the total momentum remains constant provided there is no resultant force acting on the system
- In other words, in a **closed** system, the total momentum of two bodies before a collision is equal to their total momentum after the collision

A & B are incorrect because these are definitions of force and momentum

C is incorrect because the statement is true given there are no external forces acting, it needs clarification that this is only true in a closed or isolated system

Question 2 Easy [1 mark]

The graph shows how the momentum of a motorcycle changes with time.



What is the resultant force on the motorcycle?

A 50 N

B 500 N

C 2500 N

D 5000 N

Mark Scheme & Explanation:

*The correct answer is **A** because:*

- Force is equal to the rate of change of momentum
- This is equal to the gradient of a momentum-time graph
  - Force=$Force = \frac{\Delta y}{\Delta x} = \frac{500}{10} = 50 \quad \text{N}$

Question 3 Easy [1 mark]

Two bodies travelling in a straight line collide in a perfectly elastic collision. Which of the following statements must be correct?

A the initial speed of one body will be the same as the final speed of the other body

B the relative speed of approach between the two bodies equals their relative speed of separation

C the total momentum is conserved but the total kinetic energy will be reduced

D one of the bodies will be stationary at one instant

Mark Scheme & Explanation:

*The correct answer is **B** because:*

- A perfectly elastic collision is defined as a collision in which the initial kinetic energy equals the final kinetic energy
- In other words, the relative speed of approach equals the relative speed of separation of the two bodies because kinetic energy and speed is related by
    - $E_k = \frac{1}{2}mv^2$

A & D are incorrect because these are possible elastic collision scenarios, but they do not describe all elastic collisions

C is incorrect because this describes an inelastic collision

Question 4 Easy [1 mark]

Which row correctly states whether momentum and kinetic energy are conserved in an inelastic collision in which there are no external forces?

	Momentum	Kinetic energy
A	conserved	conserved
B	not conserved	conserved
C	conserved	not conserved
D	not conserved	not conserved

A

B

C

D

Mark Scheme & Explanation:

*The correct answer is **C** because:*

- An elastic collision is a collision in which kinetic energy is conserved
- So conversely, an inelastic collision is a collision in which kinetic energy is **not** conserved
- Momentum is conserved in both elastic and inelastic collisions
- The principle of conservation of momentum states that:
    - For a system of interacting bodies, the total momentum remains constant provided there is no resultant force acting on the system

A is incorrect because the definition of an inelastic collision is that kinetic energy is not conserved as energy may be lost as sound, heat, light etc

B & D are incorrect because for any system where there are no external forces acting, momentum is always conserved

Question 5 Easy [1 mark]

Which quantity has the same base units as momentum?

A density × volume × velocity

B density × energy

C pressure × area

D weight ÷ area

Mark Scheme & Explanation:

*The correct answer is **A** because:*

- The unit of density $\rho = \frac{m}{V}$ is $\text{kg}\ \ \text{m}^{-3}$
- The unit of volume is m^3
- The unit of velocity is $\text{m}\ \ \text{s}^{-1}$
- So, the unit of density × volume × velocity is
  - $\text{kg}\ \ \text{m}^{-3}\,\text{m}^3\,\text{m}\ \ \text{s}^{-1}$
  - $\text{kgm}\ \ \text{s}^{-1}$

This is the same unit as momentum $p = mv$, which is $\text{kg}\ \ \text{m}\ \ \text{s}^{-1}$

B is incorrect because	the unit of density $\rho = \frac{m}{V}$ is $\text{kg}\ \ \text{m}^{-3}$ the unit of energy $E_k = \frac{1}{2}mv^2$ *is* $\text{J} = \text{kg}\ \ \text{m}^2\,\text{s}^{-2}$ so, the unit of density × energy is $\text{kg}\ \ \text{m}^{-3}\,\text{kg}\ \ \text{m}^2\,\text{s}^{-2} = \text{kg}^2\ \ \text{m}^{-1}\,\text{s}^{-2}$
C is incorrect because	the unit of pressure $P = \frac{F}{A}$ *is* $\text{N}\ \ \text{m}^{-2} = \text{kg}\ \ \text{m}^{-1}\,\text{s}^{-2}$ the unit of area is m^2 so, the unit of pressure × area is $\text{kg}\ \ \text{m}^{-1}\ \ \text{s}^{-2}\,\text{m}^2 = \text{kg}\ \ \text{m}\ \ \text{s}^{-2}$ (which is equal to N)

D is incorrect because	the unit of weight is $\mathrm{N} = \mathrm{kg}\ \ \mathrm{m}\ \ \mathrm{s}^{-2}$ the unit of area is m^2 so, the unit of weight ÷ area is $\mathrm{kg}\ \ \mathrm{m}\ \ \mathrm{s}^{-2} / \mathrm{m}^2 = \mathrm{kg}\ \ \mathrm{m}^{-1}\,\mathrm{s}^{-2}$ (which is equal to Pa)

Question 6 Easy [1 mark]

Which of the following is a statement of the principle of conservation of momentum?

A in an elastic collision momentum is constant

B momentum is the product of mass and velocity

C the force acting on a body is proportional to its rate of change of momentum

D the momentum of an isolated system is constant

Mark Scheme & Explanation:

*The correct answer is **D** because:*

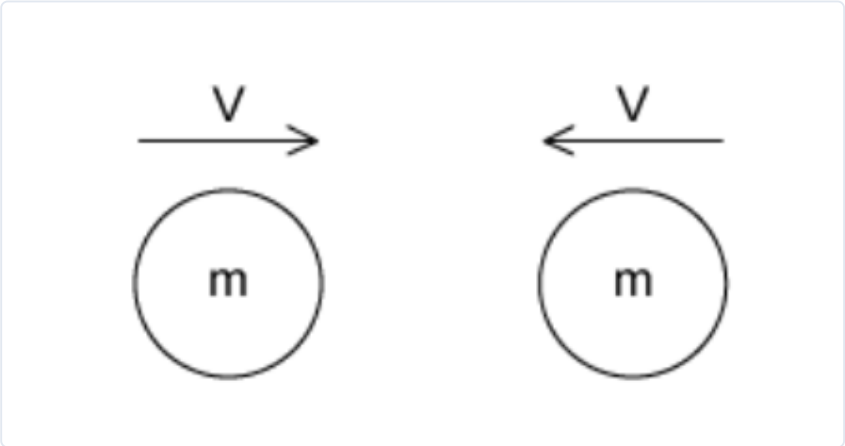
- The principle of conservation of momentum states that:
  - For a system of interacting bodies, the total momentum remains constant provided there is no resultant force acting on the system
- In other words, in a **closed** or **isolated** system, the total momentum of two bodies before a collision is equal to their total momentum after the collision

A is incorrect because this is true, however it needs to clarify that momentum is only constant in a closed or isolated system

B & C are incorrect because these are definitions of force and momentum

Question 7 Easy [1 mark]

Two similar spheres, each of mass *m* and travelling with speed *v*, are moving towards each other



The spheres have a head-on elastic collision

Which statement is correct?

A the spheres stick together on impact

B the total kinetic energy after impact is mv^2

C the total kinetic energy before impact is zero

D the total momentum before impact is $2mv$

Mark Scheme & Explanation:

*The correct answer is **B** because:*

- Kinetic energy $=\frac{1}{2}mv^2$
- Each sphere moves with speed v before the impact, so the kinetic energy of each sphere $=\frac{1}{2}mv^2$
- So, the sum of the kinetic energies is $=\frac{1}{2}mv^2 + \frac{1}{2}mv^2 = mv^2$
- The spheres collide elastically, meaning no kinetic energy is lost so the kinetic energy before the impact = kinetic energy after the impact = mv^2

A is incorrect because this is the definition of a perfectly inelastic collision

C is incorrect because the total kinetic energy before impact would be zero if energy was a vector quantity, but it is a scalar so the total kinetic energy is the sum of the energies

D is incorrect because momentum is a vector, because the spheres are moving in opposite directions the total momentum is equal to mv – mv = 0

Question 8 Easy [1 mark]

Trolley X, moving along a horizontal frictionless track, collides with a stationary trolley Y. The two trolleys become attached and move off together.

Which statement about this interaction is correct?

A some of the kinetic energy of trolley X is changed to momentum in the collision

B some of the momentum of trolley X is changed to kinetic energy in the collision

C trolley X loses some of its momentum as heat in the collision

D trolley X shares its momentum with trolley Y but some kinetic energy is lost

Mark Scheme & Explanation:

*The correct answer is **D** because:*

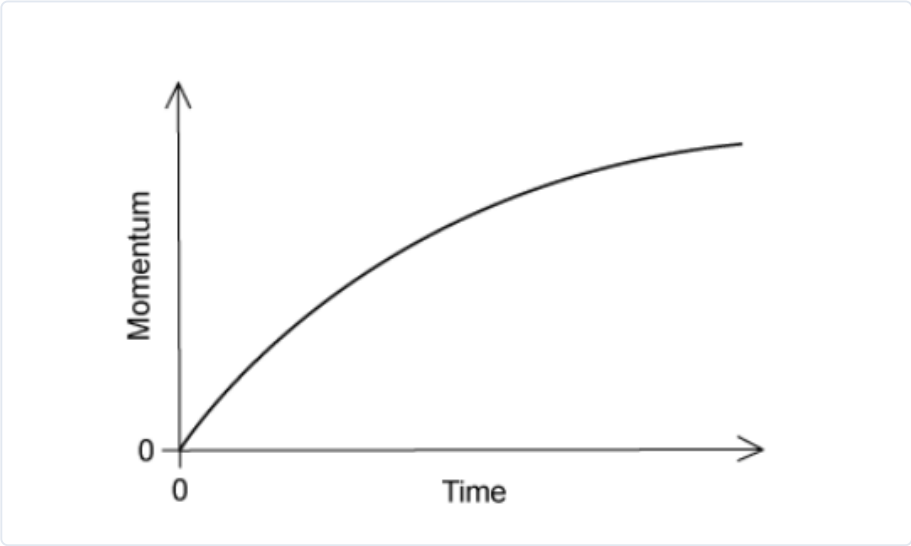
- A collision in which the objects stick together and have the same final velocity is called a perfectly inelastic collision
- This is because it reduces the initial kinetic energy more than any other type of inelastic collision
    - Option D is therefore correct because we are told the trolleys "become attached and move off together" indicating it is an inelastic collision and some kinetic energy is lost as a result
- The principle of conservation of momentum states that (in a closed system) the total momentum of the two trolleys before the collision is equal to their total momentum after the collision
    - Option D is therefore correct because there are no other bodies involved apart from the trolleys, so momentum must be conserved

A & B are incorrect because momentum cannot be "changed" into kinetic energy, and vice versa

C is incorrect because momentum cannot be "lost" via heat, as heat is a form of energy and momentum is conserved in isolated systems

Question 9 Easy [1 mark]

A car accelerates from rest. The graph shows the momentum of the car plotted against time.



What is the meaning of the gradient of the graph at a particular time?

A the velocity of the car

B the resultant force on the car

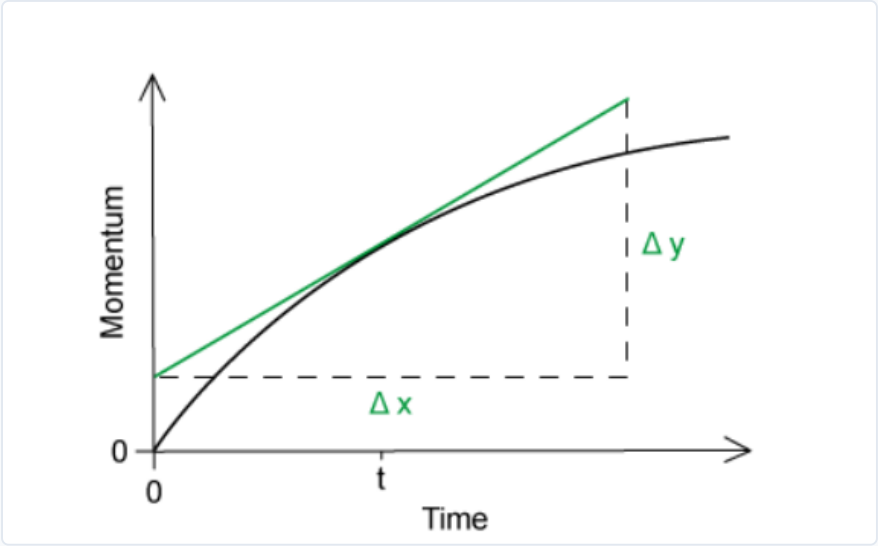
C the kinetic energy of the car

D the rate of change of kinetic energy of the car

Mark Scheme & Explanation:

*The correct answer is **B** because:*

- The gradient of a momentum-time graph can be found at a particular time, *t*, by drawing a tangent on the curve at *t* then using
  - $\frac{\Delta y}{\Delta x} = \frac{\Delta p}{\Delta t}$
- Force is equal to the rate of change of momentum so the gradient is equal to the resultant force on the car



A is incorrect because momentum is equal to p = mv, so velocity would be the gradient of a momentum-mass graph

C is incorrect because kinetic energy is equal to $E_k = \frac{1}{2}mv^2 = \frac{1}{2}pv$, so kinetic energy would be the area under a momentum-velocity graph

D is incorrect because the rate of change of kinetic energy is equal to power

Question 10 Easy [1 mark]

Which statement about a ball that strikes a tennis racket and rebounds is **always** correct?

A total kinetic energy of the ball is conserved

B total kinetic energy of the system is conserved

C total momentum of the ball is conserved

D total momentum of the system is conserved

Mark Scheme & Explanation:

*The correct answer is **D** because:*

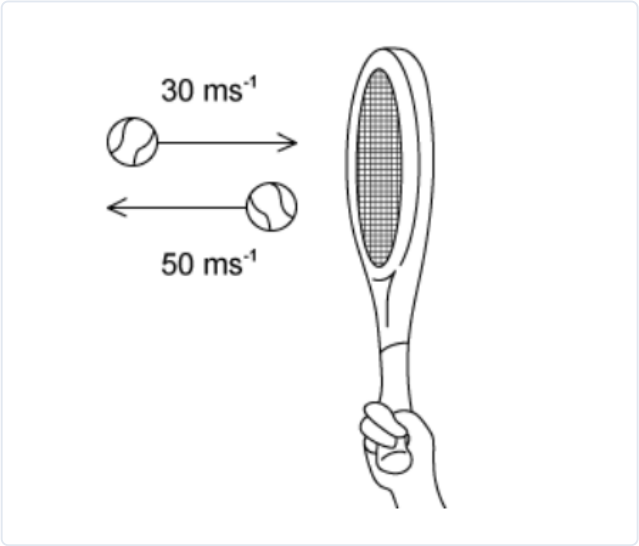
- The principle of conservation of momentum tells us that the momentum of a closed system is always conserved (provided no external forces act)

A & B are incorrect because it isn't possible to say that kinetic energy will **always** be conserved in a collision because the energy can be transferred to heat, light or sound

C is incorrect because when the tennis ball is struck by the racket, the velocity of the ball will increase, so the momentum of the individual ball will increase but the momentum of the racket will decrease to ensure the momentum of the system is conserved

Question 11 Medium [1 mark]

A tennis ball of mass 60 g is struck by a tennis racket. The velocity of the ball changes as shown



What is the magnitude of the change in momentum of the ball?

A 1.2 kg m s^{-1}

B 4.8 kg m s^{-1}

C 1200 kg m s^{-1}

D 4800 kg m s^{-1}

Mark Scheme & Explanation:

*The correct answer is **B** because:*

- The ball changes direction after it is hit by the racket
- Take the positive direction as the direction of the ball after the collision, so:
  - Initial velocity, $v_i = -30 \text{ m s}^{-1}$
  - Final velocity, $v_f = +50 \text{ m s}^{-1}$
- Change in momentum is equal to
  - $\Delta p = m\Delta v$
  - $\Delta p = mv_f - v_i$
  - $\Delta p = 60 \times 10^{-3} \times 50 - -30$

- $\Delta p = 4.8 \text{ kg m s}^{-1}$

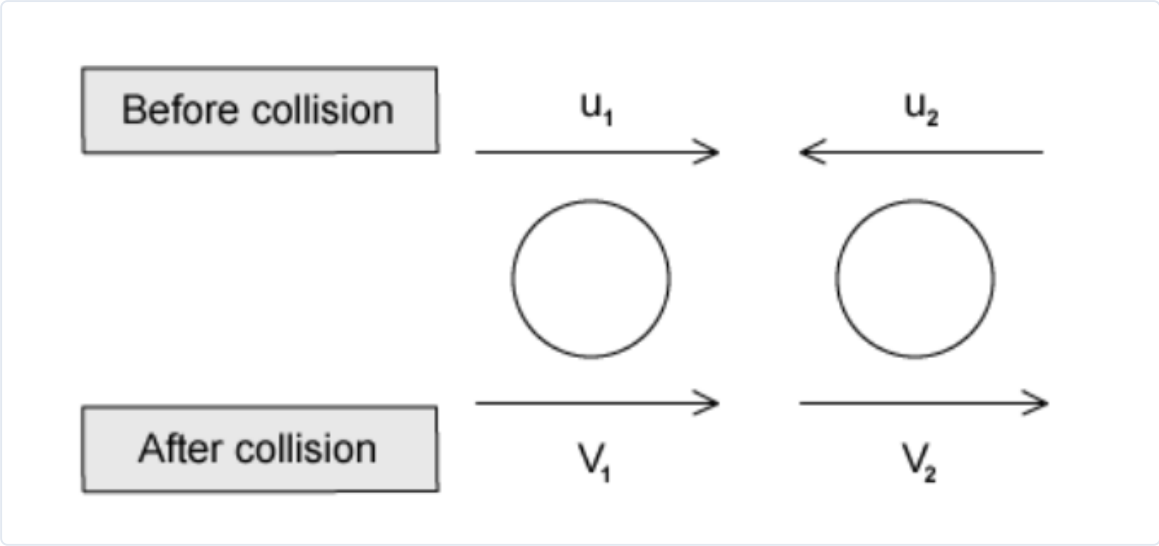
A is incorrect as this is the answer you get if you don't set a positive and negative direction and add the velocities together.

C is incorrect as this is the answer you get if you don't convert the mass from g to kg, and if you don't set a positive and negative direction and add the velocities together.

D is incorrect as this is the answer you get if you don't convert the mass from g to kg.

Question 12 Medium [1 mark]

Two spheres approach each other along the same straight line. Their speeds are u_1 and u_2 before collision, and v_1 and v_2 after collision, in the directions shown below



Which equation is correct if the collision is perfectly elastic?

A $u_1 - u_2 = v_2 + v_1$

B $u_1 - u_2 = v_2 - v_1$

C $u_1 + u_2 = v_2 - v_1$

D $u_1 + u_2 = v_2 + v_1$

Mark Scheme & Explanation:

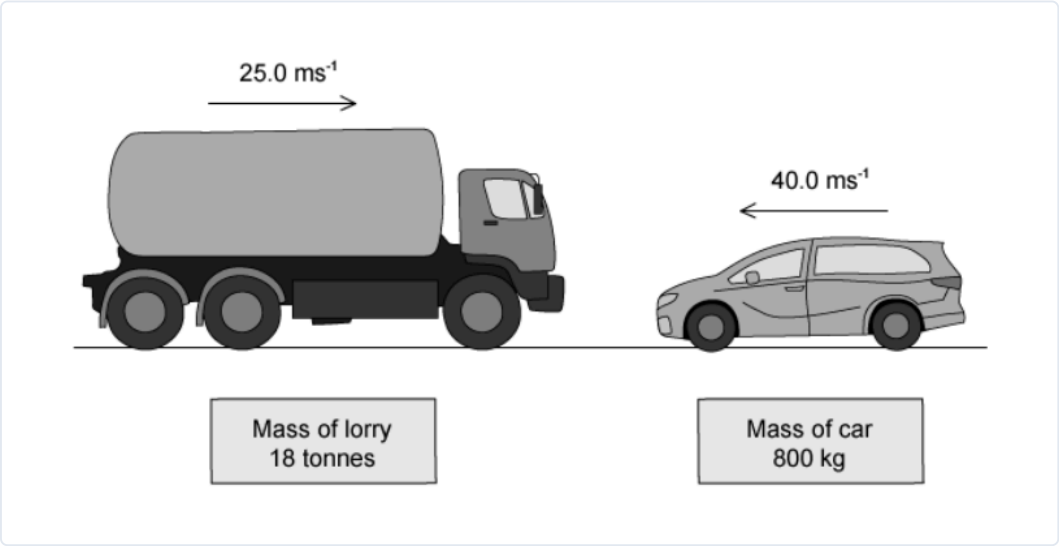
*The correct answer is **C** because:*

- In a perfectly elastic collision, **both** momentum and kinetic energy are conserved
- In an elastic collision: the relative speed of approach before the collision must be equal to the relative speed of separation after the collision
- Before the collision: the two spheres are moving towards each other, they are approaching each other
    - The relative speed of approach is $u_1 + u_2$
- After the collision: the spheres move in the same direction, the first sphere is moving towards the second sphere and the second sphere is moving away from the first – this is an important distinction to make as it tells us v_2 must be greater than v_1

    - The relative speed of separation is v_2 - v_1
- So, the relative speeds before and after the collision are equal to
    - $u_1 + u_2 = v_2$ - v_1

Question 13 Medium [1 mark]

A lorry of mass 18 tones is travelling at 25.0 $m \quad s^{-1}$



A car of mass 800 kg is travelling at 40.0 $m \quad s^{-1}$ towards the lorry.

What is the magnitude of the total momentum?

A 282 kN s

B 418 kN s

C 482 kN s

D 1222 kN s

Mark Scheme & Explanation:

*The correct answer is **B** because:*

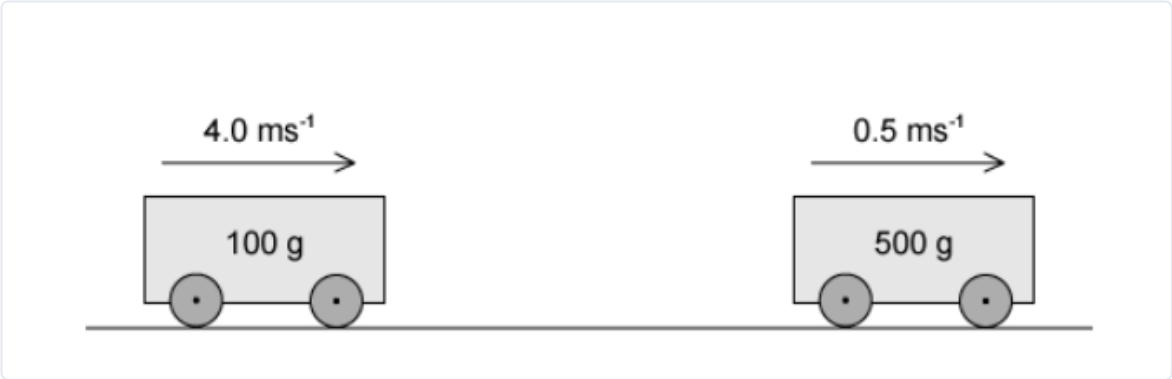
- Momentum is a vector quantity, so we need to set out a positive and negative direction
- Let's define the positive direction as the direction towards the right
- Momentum is equal to $p = mv$
- Total momentum = Momentum of lorry + Momentum of car
- Total momentum = (18000 × 25) + (800 × – 40)

  = 418000 N s

= **418 kN s**

Question 14 Medium [1 mark]

The diagram shows the masses and velocities of two trolleys about to collide



After the impact they move off together

What is the total kinetic energy of the trolleys after the collision?

A 6.0 J

B 3.7 J

C 0.35 J

D 0.02 J

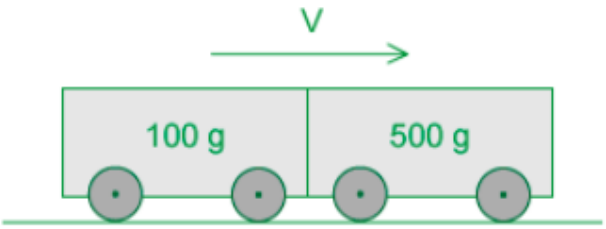
Mark Scheme & Explanation:

*The correct answer is **C** because:*

- After the collision, the trolleys move together as a single trolley of mass 600 g and velocity *v*
- We can find the value of *v* using the principle of conservation of momentum:
  - The sum of momentum before the collision equals the sum of momentum after the collision
- Momentum before: $100 \times 4 + 500 \times 0.5 = 400 + 250 = 650$
  - Note: this is the same as writing

    $100 \times 10^{-3} \times 4 + 500 \times 10^{-3} \times 0.5 = 650 \times 10^{-3}$
- Momentum after: $100 + 500v = 600v$
- Therefore, $650 = 600 \quad v$
  - $v = \frac{650}{600} = 1.08 \quad \text{m} \quad \text{s}^{-1}$

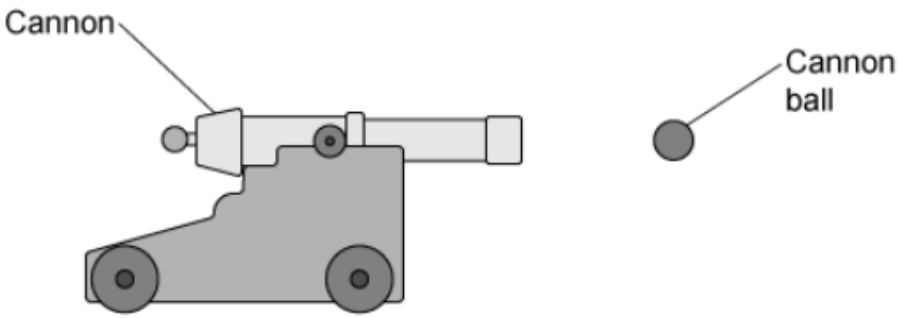
- Kinetic energy $E_k = \frac{1}{2}mv^2$
  - $E_k = \frac{1}{2}mv^2 = \frac{1}{2} \times 600 \times 10^{-3} \times 1.08^2 = 0.35J$



Question 15 Medium [1 mark]

The diagram shows a cannon ball fired from a cannon

The mass of the cannon is 1600 kg and the mass of the cannon ball is 20 kg



The recoil velocity of the cannon is 5 m s^{-1} horizontally

What is the horizontal velocity of the cannon ball?

A 100 m s^{-1}

B 320 m s^{-1}

C 400 m s^{-1}

D 8000 m s^{-1}

Mark Scheme & Explanation:

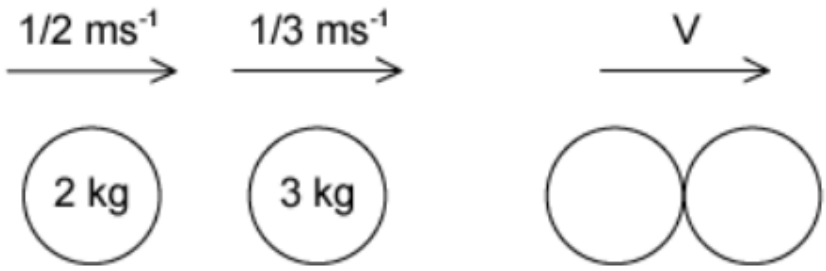
*The correct answer is **C** because:*

- Before the cannon ball is fired both the cannon and the ball are stationary so the initial momentum is zero
- The cannon recoils in the opposite direction to the direction of motion of the cannon ball, so we must define one of the directions to be positive and one to be negative
- By the principle of conservation of momentum, the sum of momentum before the collision equals the sum of momentum after the collision
  - $0 = 1600 \times 5 - 20 \quad v$
  - $20 \quad v = 8000$

- $v = 400 \quad \text{m} \quad \text{s}^{-1}$

Question 16 Medium [1 mark]

A ball of mass 2 kg travelling at ½ m s^{-1} strikes a ball of mass 3 kg travelling at ⅓ m s^{-1}. Both balls are moving along the same straight line as shown



After collision, both balls move at the same velocity *v*

What is the magnitude of the velocity *v*?

A $\frac{1}{5}$ m s^{-1}

B $\frac{2}{5}$ m s^{-1}

C $\frac{3}{2}$ m s^{-1}

D $\frac{5}{2}$ m s^{-1}

Mark Scheme & Explanation:

*The correct answer is **B** because:*

- After the collision, the balls move together as a single object of mass 5 kg and velocity *v*
- We can find the value of *v* using the principle of conservation of momentum:
- The sum of the momentum before the collision equals the sum of the momentum after the collision
  - $2 \times \frac{1}{2} + 3 \times \frac{1}{3} = 5 \quad v$
  - $5v \quad = \quad 2$
  - $v = \quad \frac{2}{5}$ m s^{-1}

Question 17 Medium [1 mark]

A molecule of mass *m* travelling at speed *v* hits a wall in a direction perpendicular to the wall. The collision is elastic.

What are the changes in the momentum and in the kinetic energy of the molecule caused by the collision?

	Change in momentum	Change in kinetic energy
A	0	0
B	mv	mv^2
C	mv^2	0
D	$2mv$	0

A

B

C

D

Mark Scheme & Explanation:

*The correct answer is **D** because:*

- For any collision in a closed system, momentum is always conserved
- Before the collision:
  - Momentum of the molecule = mv
- After the collision:
  - The molecule will still have the same momentum but it will move in the opposite direction so the momentum of the molecule = $-mv$
- Change in momentum = mv - $-mv = 2 \quad mv$
- The collision is elastic so the kinetic energy is conserved, therefore, the change in kinetic energy = **0**

A is incorrect because the change in momentum cannot be zero because the molecule changes direction

B is incorrect because the change in kinetic energy must be zero because the collision is elastic meaning the kinetic energy is conserved and does not change

C is incorrect because mv^2 has the units of energy

Question 18 Medium [1 mark]

A moving object strikes a stationary object. The collision is inelastic. The objects move off together.

Which row shows the possible values of total momentum and total kinetic energy for the system before and after the collision?

	Total momentum before collision / kg m s^{-1}	Total momentum after collision / kg m s^{-1}	Total kinetic energy before collision / J	Total kinetic energy after collision / J
A	6	6	90	30
B	6	6	30	90
C	6	2	90	30
D	6	6	90	90

A

B

C

D

Mark Scheme & Explanation:

*The correct answer is **A** because:*

- For any collision in a closed system, momentum is always conserved, so the total momentum before and after the collision should be the same
- The collision is inelastic, meaning that kinetic energy is not conserved, so the kinetic energy after the collision should be less than the energy before

B is incorrect because the kinetic energy before the collision cannot be lower than the kinetic energy after the collision

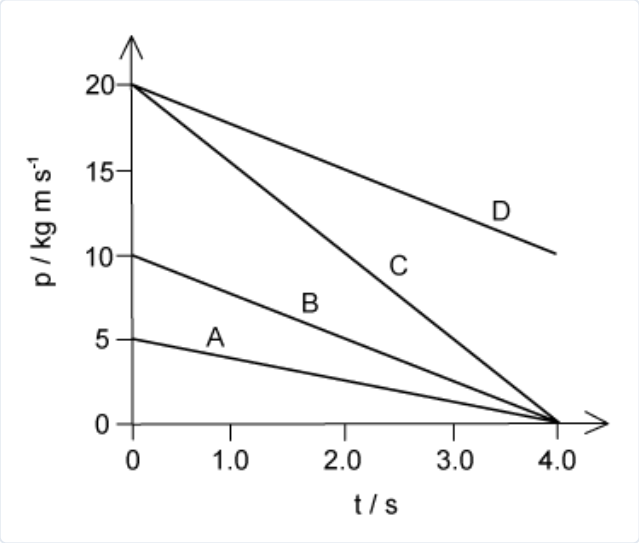
C is incorrect because momentum is always conserved, so the momentum should be the same before and after the collision

D is incorrect because the collision is inelastic, meaning the energy is not conserved, so the energy should be less than the initial energy

Question 19 Medium [1 mark]

A resultant force of 5 N acts on a body for a time of 4.0 s

Which graph could show the variation with time *t* of the momentum *p* of the body?



A

B

C

D

Mark Scheme & Explanation:

*The correct answer is **C** because:*

- Force is equal to the rate of change of momentum
- In other words, force is equal to the gradient $\frac{\Delta y}{\Delta x}$ of a momentum-time graph
  - $F = \frac{\Delta p}{\Delta t}$
- The gradient of the line should be equal to 5 N
- The gradient of line C is $\frac{\Delta y}{\Delta x} = \frac{20}{4} = 5 \quad \text{N}$

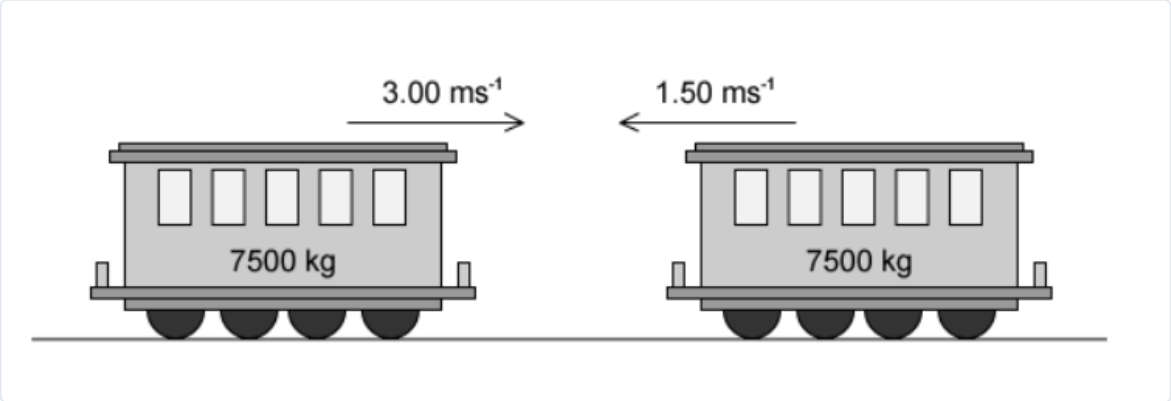
A is incorrect because the gradient of this line is $\frac{5}{4} = 1.25 \quad \text{N}$

B is incorrect because the gradient of this line is $\frac{10}{4} = 2.5 \quad \text{N}$

D is incorrect because the gradient of this line is $\frac{20 - 10}{4} = 2.5 \quad \text{N}$

Question 20 Medium [1 mark]

Two train carriages each of mass 7500 kg roll toward one another on a level track. One is travelling at 3.00 m s^{-1} and the other at 1.50 m s^{-1} as shown.



They collide and join together.

What is the kinetic energy lost during the collision?

A 4200 J

B 7500 J

C 11 250 J

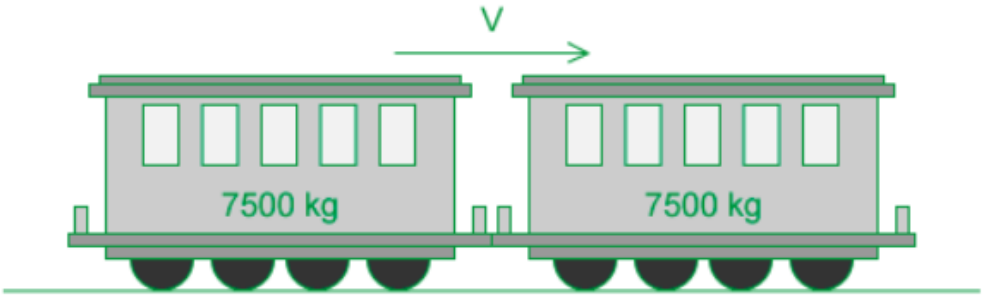
D 38 000 J

Mark Scheme & Explanation:

*The correct answer is **D** because:*

- After the collision, the train carriages move together as a single object of mass 15 000 kg and velocity *v*
- We can find the value of *v* using the principle of conservation of momentum:
  - The sum of the momentum before the collision equals the sum of the momentum after the collision
- Let's define the positive direction to be to the right
- Momentum before: $7500 \times 3 - 7500 \times 1.5 = 22\ 500 - 11\ 250 = 11\ 250$
- Momentum after: $7500 + 7500v = 15000\ v$
  - $11\ 250 = 15\ 000v$
  - $v = \frac{11\ 250}{15\ 000} = 0.75\ \text{m s}^{-1}$

- Kinetic energy $E_k = \frac{1}{2}mv^2$ before collision = sum of kinetic energies of the train carriages
- Kinetic energy before collision = $\frac{1}{2}75003^2 + \frac{1}{2}75001 . 5^2 = 42 \quad 200 \quad \text{J}$
- Kinetic energy after collision = $\frac{1}{2}15 \quad 0000 . 75^2 = 4200 \quad \text{J}$
    - Kinetic energy lost = $42 \quad 200 - 4200 = 38000 \quad \text{J}$



Question 21 Medium [1 mark]

A spacecraft of mass 1000 kg and a fuel module of mass m are initially travelling together at 13 m s^{-1} in a straight line.

The fuel module separates from the rocket and continues to travel along the same line at 5 m s^{-1} while the spacecraft continues to travel along the same line at 17 m s^{-1}.

What is the mass m of the fuel module?

A 300 kg

B 500 kg

C 800 kg

D 1000 kg

Mark Scheme & Explanation:

*The correct answer is **B** because:*

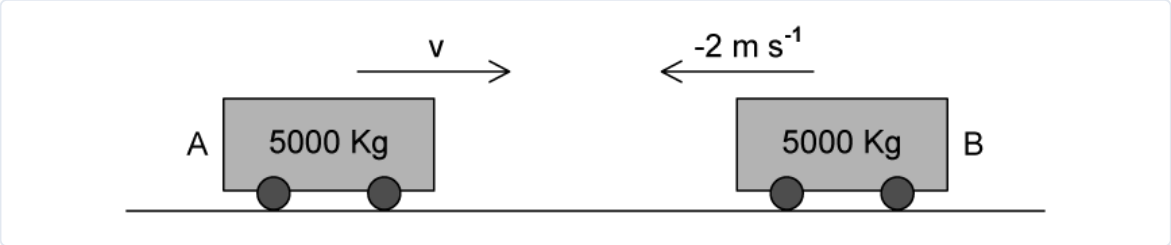
- Momentum: $p = mv$
- Before separation:
  - Combined mass = $1000 + m$
  - Speed = 13 m s^{-1}
- The initial momentum of the spacecraft and fuel module is:
  - $p_i = 1000 + m \times 13$
  - $p_i = 13\,000 + 13m$
- After separation:
  - Spacecraft momentum = $1000 \times 17 = 17\,000 \text{ kg m s}^{-1}$
  - Fuel module momentum = $5m$
- All values of momentum are positive as the spacecraft and module continue to travel in the same direction
- The final momentum of the spacecraft and fuel module is:
  - $p_f = 1000 \times 17 + 5m$
  - $p_f = 17\,000 + 5m$
- Apply the principle of conservation of momentum:

- $p_i = p_f$
- $13\ 000 + 13m = 17\ 000 + 5m$
- $8m = 4000$
- $m = \frac{4000}{8} = 500\ \text{kg}$

Even though there is no collision, the principle of conservation of momentum still applies as no external forces act on the system. However, if, for example, the spacecraft were burning fuel, then this would be considered an external force, and momentum would not be conserved.

Question 22 Medium [1 mark]

Two trolleys, A and B, each have a mass of 5000 kg. Trolley A travels at speed v towards trolley B. Trolley B travels at 2.0 m s^{-1} in the opposite direction, as shown.



The trolleys collide and join together with a combined kinetic energy of 1250 J.

What is the velocity of trolley A before the collision?

A −1.0 m s^{-1}

B 3.0 m s^{-1}

C 3.4 m s^{-1}

D 4.0 m s^{-1}

Mark Scheme & Explanation:

*The correct answer is **B** because:*

- Calculate the final velocity of the connected trolleys:
  - $E_k = \frac{1}{2}mv_f^2 = 1250$ J
  - $v_f = \sqrt{\frac{2\times 1250}{m}}$
  - $v_f = \sqrt{\frac{2\times 1250}{5000+5000}} = 0.5$ m s^{-1}
- Calculate the total momentum before the collision:
  - $p_i = m_A u_A + m_B u_B$
  - $p_i = 5000v - 2\times 5000$
  - $p_i = 5000v - 10\ 000$
- Calculate the total momentum after the collision:
  - $p_f = m_A + m_B v_f$

- $p_f = 5000 + 5000 \times 0.5$
- $p_f = 5000 \text{ kg m s}^{-1}$

- Apply the principle of conservation of momentum:
  - $p_i = p_f$
  - $5000v - 10\,000 = 5000$
  - $v = \frac{15\,000}{5000} = 3.0 \text{ m s}^{-1}$

A is incorrect because this is the velocity if a minus sign is not used to take into account the direction of trolley B.

C is incorrect because this is the velocity if the masses of both trolleys are not added together in the calculation of momentum after the collision

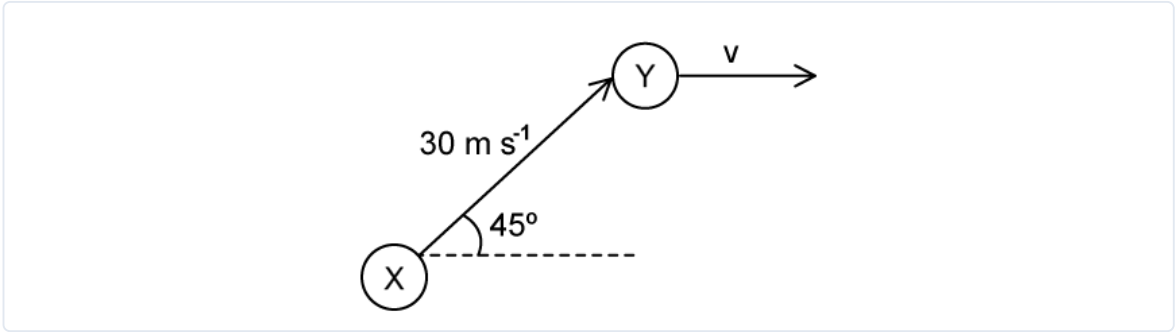
D is incorrect because this is the answer if multiplication by velocity is omitted from the calculation of momentum after the collision

Question 23 Medium [1 mark]

Two identical billiard balls of mass m are placed on a smooth horizontal surface.

Ball Y is initially at rest. Ball X approaches ball Y with a speed of 30 m s^{-1} at an angle of 45° to the direction shown.

The balls collide perfectly elastically. After the collision, ball Y moves with speed v, and ball X moves at right angles to the direction of motion of ball Y.



What is the speed v of ball Y after the collision?

A 21 m s^{-1}

B 26 m s^{-1}

C 30 m s^{-1}

D 39 m s^{-1}

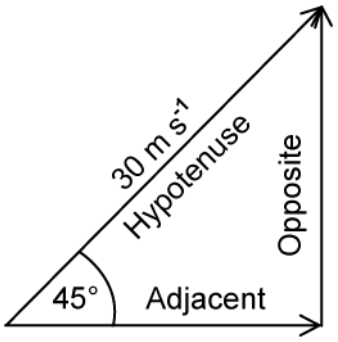
Mark Scheme & Explanation:

*The correct answer is **A** because:*

- For collisions in two dimensions, momentum is conserved independently along each perpendicular direction
- Since the balls have identical masses and have a perfectly elastic collision, they move at 90° to one another afterwards, which means
  - ball Y moves in the horizontal direction
  - ball X moves in the vertical direction
- In the horizontal direction:
  - Before the collision: $p_X = m30 \cos 45°$
  - After the collision: $p_Y = mv$
- Apply the principle of conservation of momentum in the horizontal direction:

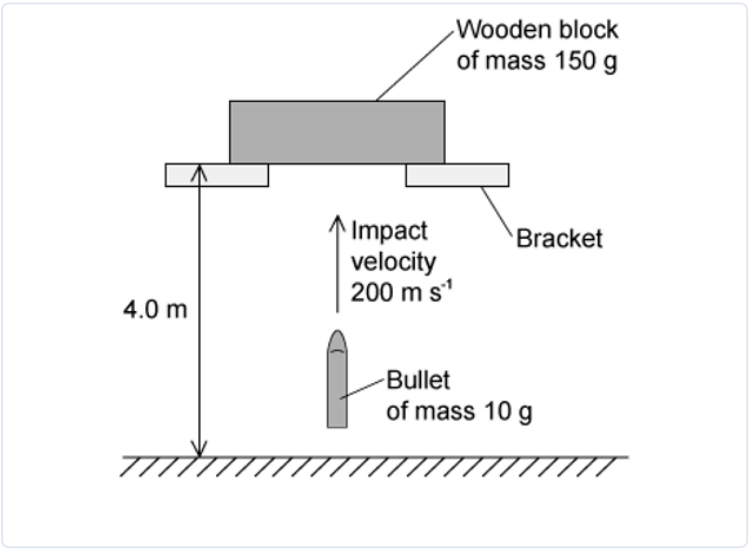
- $p_i = p_f$
- $p_X + 0 = 0 + p_Y$
- $m30 \cos 45° = mv$
- $v = 30 \cos 45° \approx 21 \text{ m s}^{-1}$

To determine the horizontal component of ball X's velocity using trigonometry, remember SOH CAH TOA:



Question 24 Hard [1 mark]

A wooden block is freely supported on brackets at a height of 4.0 m above the ground, as shown



A bullet of mass 10.0 g is shot vertically upwards into the wooden block of mass 150 g. It embeds itself in the block. The impact causes the block to rise above its supporting brackets.

The bullet hits the block with a velocity of 200 m s^{-1}. How far above the ground will the block be at the maximum height of its path?

A 8.0 m

B 10.0 m

C 12.0 m

D 14.0 m

Mark Scheme & Explanation:

*The correct answer is **C** because:*

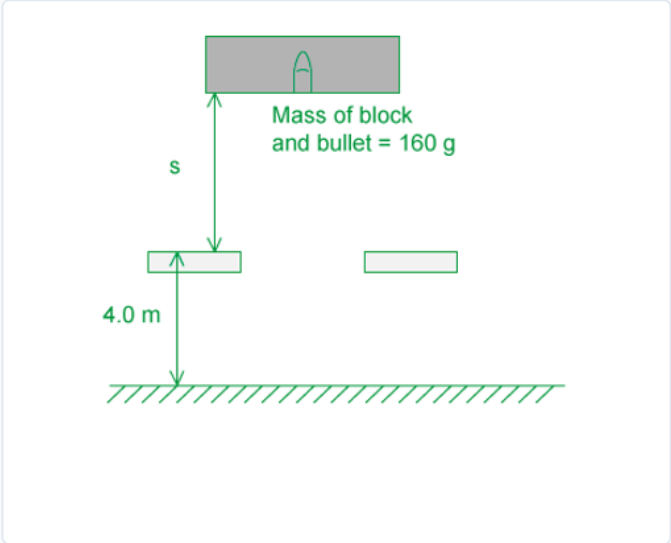
Part 1: use the conservation of momentum to find the speed after collision

- Sum of momentum before collision = Sum of momentum after collision
  - Momentum of block before collision $= 0$
  - Momentum of bullet before collision $= 10g \times 200$ m s^{-1}
- The bullet becomes embedded in the block, so the two objects will travel together with the same velocity, *u*

- The momentum after the collision $= 10 \quad g + 150 \quad gu = 160 \quad u$
- $160 \quad u = 2000$
- $u = 12.5 \quad \text{m} \quad \text{s}^{-1}$

Part 2: use a kinematic equation to find the maximum height

- List the quantities we know:
  - u = 12.5 m s^{-1}
  - v = 0
  - a = –9.81 m s^{-2}
  - s = ?
- The equation that links *u, v, a* and *s* is
  - $v^2 = u^2 + 2as$
  - $0 = 12.5^2 + 2\text{-}9.81s$
  - $2 \times 9.81s = 12.5^2$
  - $s = \frac{12.5^2}{19.6} = 8.0 \quad \text{m}$
- Height of the block above the ground = 4.0 + 8.0 = 12.0 m



Question 25 Hard [1 mark]

Two railway trucks of masses *m* and 5 *m* move towards each other in opposite directions with speeds 3 *u* and *u* respectively. These trucks collide and stick together.

What is the speed of the trucks after the collision?

A $\frac{u}{6}$

B $\frac{3u}{6}$

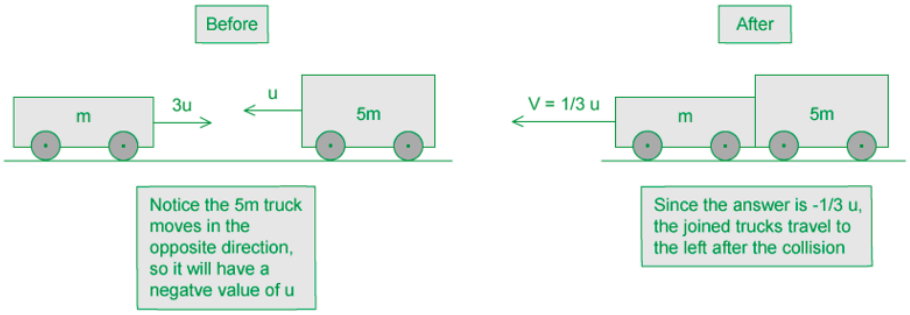
C $\frac{u}{3}$

D $\frac{5u}{3}$

Mark Scheme & Explanation:

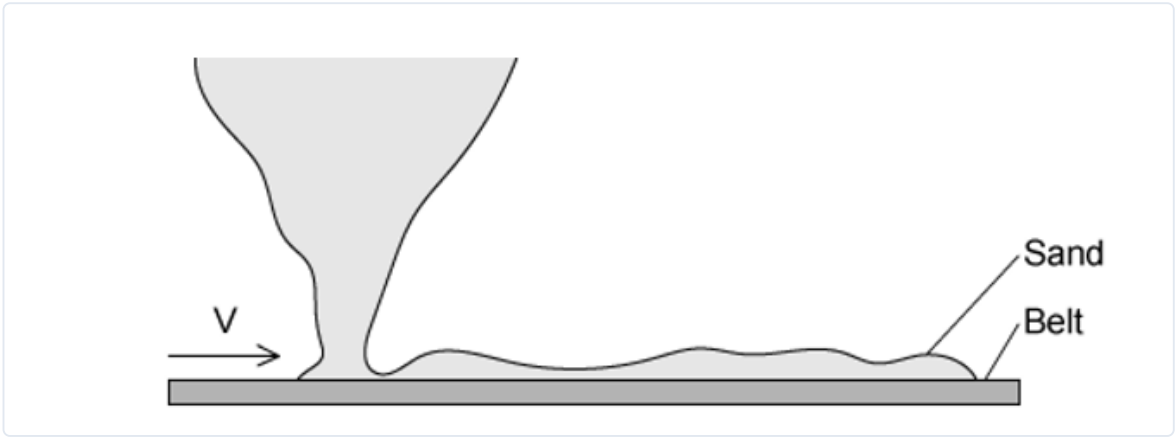
*The correct answer is **C** because:*

- Using the principle of conservation of momentum:
  - Sum of momentum before collision = Sum of momentum after collision
- Let's set the positive direction to be to the right
- Momentum of trucks before collision
  - $m \times 3u + 5m \times -u = 3mu - 5mu = -2mu$
- After the collision the trucks join together the total mass becomes 6*m* and the combined trucks move at a final speed *v*
- Momentum of trucks after collision = $m + 5mv = 6mv$
  - $-2mu = 6mv$
  - $u = -3v$
  - $v = -\frac{1}{3}u$
- Speed does not have a direction so the answer is $\frac{u}{3}$


Before
m
3u
u
5m
Notice the 5m truck moves in the opposite direction, so it will have a negatve value of u
After
V = 1/3 u
m
5m
Since the answer is -1/3 u, the joined trucks travel to the left after the collision

Question 26 Hard [1 mark]

Sand falls vertically on a conveyor belt at a rate of m kg s^{-1}



In order to keep the belt moving at constant speed v, the horizontal force that must be exerted is equal to

A mv

B $\frac{1}{2} mv$

C mv^2

D $\frac{1}{2} mv^2$

Mark Scheme & Explanation:

*The correct answer is **A** because:*

- Usually when we deal with momentum questions it is the velocity that is changing, but in this case, the velocity is constant and it is the mass of the sand that is changing at a rate of m kg s^{-1} (m here means mass, not metres)
- We can define the rate of change of mass as
  - $m = \frac{\Delta M}{t}$
- So, the momentum is equal to
  - $\Delta p = \Delta M v$
- Force is equal to the rate of change of momentum
  - $F = \frac{\Delta p}{t} = \frac{\Delta M v}{t} = \frac{\Delta M}{t} v = mv$
- We can check the units to ensure it is consistent

- The unit of force is the newton: N = kg m s^{-2}
- *m* has units of kg s^{-1}, *v* has units of m s^{-1}
- So, *mv* also has units of [kg s^{-1}] [m s^{-1}] = kg m s^{-2}

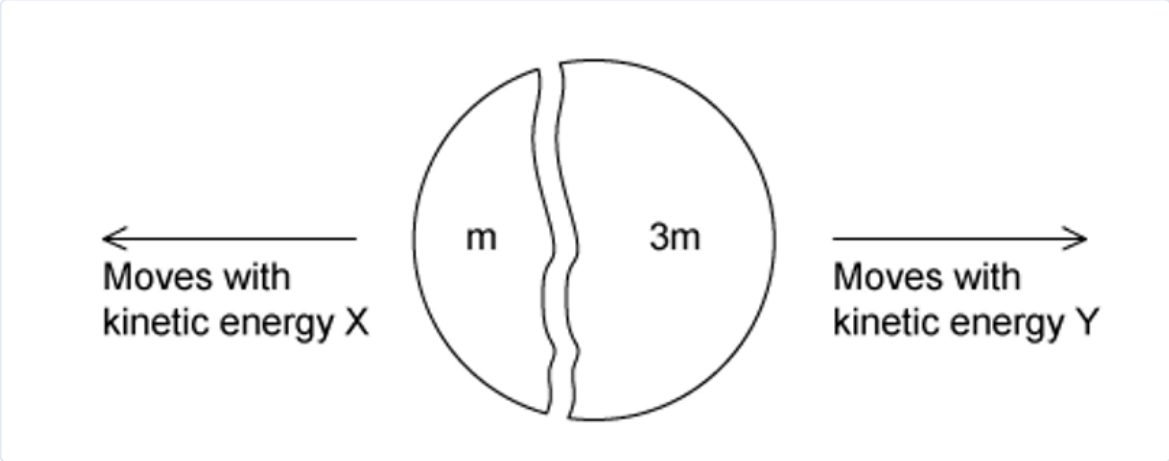
B is incorrect because the units are consistent but the factor of ½ is incorrect

C & D are incorrect because the units are inconsistent with the units of force (kg m s^{-2}) mv^2 = [kg s^{-1}] [m s^{-1}]^2 = kg m^2 s^{-3}

Question 27 Hard [1 mark]

A stationary body explodes into two components of masses *m* and 3 *m*.

The components gain kinetic energies X and Y respectively.



What is the value of the ratio $\frac{X}{Y}$?

A $\frac{6}{1}$

B $\frac{3}{1}$

C $\frac{1}{3}$

D $\frac{1}{6}$

Mark Scheme & Explanation:

*The correct answer is **B** because:*

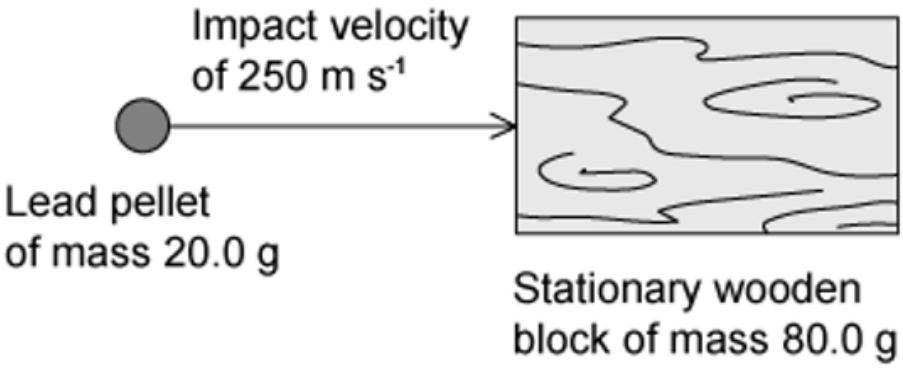
- From the conservation of momentum, the sum of the momentum before the explosion should equal the sum of momentum after the explosion
  - Momentum before explosion = 0
  - Momentum after explosion =p_{3m} - p_m
  - p_{3m} - $p_m = 0$
  - Therefore, $p_{3m} = p_m$
- This means that the magnitudes of the momentum of both components are equal
- Let's say the velocity of mass 3 *m* is *v* and the velocity of mass *m* is *v'*

- Momentum of component 3 m : $p_{3m} = 3mv$
- Momentum of component m : $p_m = mv'$
- $3mv = mv'$
- $v' = \frac{3mv}{m} = 3v$

- Kinetic energy = ½ m v^2
  - $X = \frac{1}{2}m3v^2 = \frac{9}{2}mv^2$
  - $Y = \frac{1}{2}3mv^2 = \frac{3}{2}mv^2$
- Ratio $\frac{X}{Y} = \frac{\frac{9}{2}mv^2}{\frac{3}{2}mv^2} = 3$

Question 28 Hard [1 mark]

A lead pellet of mass 20.0 g is shot horizontally into a stationary wooden block of mass 80.0 g. The pellet hits the block with an impact velocity of 250 m s^{-1}. It embeds itself in the block and it does not emerge.



After the impact, the block and the pellet slide along a rough horizontal surface and come to rest in 8.4 s.

How far do the block and the pellet travel before coming to rest?

A 175 m

B 210 m

C 260 m

D 525 m

Mark Scheme & Explanation:

*The correct answer is **B** because:*

Part 1: use the conservation of momentum to find the speed after collision

- From the conservation of momentum, the sum of the momentum before the collision should equal the sum of momentum after the collision
  - Momentum of wooden before the collision = 0
  - Momentum of pellet before the collision = $20 \quad g \times 250 \quad \text{m} \quad \text{s}^{-1}$
- After the collision the pellet becomes embedded in the wooden block, so the two travel together with a mass of (20 g + 80 g) = 100 g and a speed of *u*
  - Momentum of block and pellet after the collision = 100 *u*

- Sum of momentum before the collision = Sum of momentum after the collision
  - $20 \times 250 = 100 \quad u$
  - $u = \frac{5000}{100} = 50 \quad \text{m} \quad \text{s}^{-1}$



Part 2: use a kinematic equation to find the maximum distance

- List the quantities we know:
  - u = 50 m s^{-1}
  - v = 0
  - t = 8.4 s
  - s = ?
- The equation that links *u, v, t* and *s* is
  - $s = \frac{u + vt}{2}$
  - $s = \frac{50}{2} \times 8.4$
  - $s = 210 \quad \text{m}$
- The block and pellet will travel 210 m before coming to a stop

Question 29 Hard [1 mark]

A stationary nucleus has nucleon number *A*.

The nucleus decays by emitting a proton with speed *v* to form a new nucleus with speed *u*. The new nucleus and the proton move away from one another in opposite directions.

Which equation gives in terms of *A* and *u*?

A $v = \frac{A}{4} - 1 \quad u$

B $v = A - 1 \quad u$

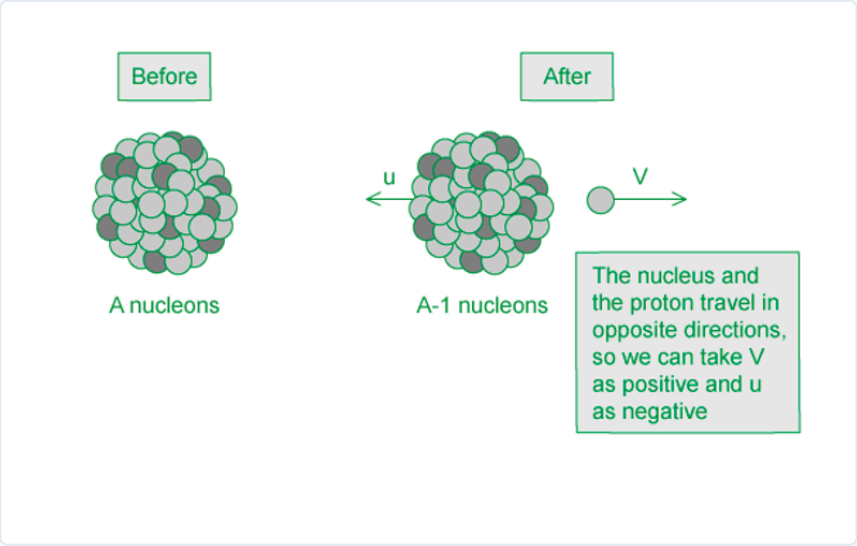
C $v = Au$

D $v = A + 1 \quad u$

Mark Scheme & Explanation:

*The correct answer is **B** because:*

- According to conservation of momentum
- Sum of momentum before decay = Sum of momentum after decay
- The nucleus is initially stationary, its speed is zero, therefore, its momentum before the decay is also initially zero
- The quantities we will need to work out the momentum after the decay are:
  - The nucleus contains *A* nucleons
  - The proton has a speed of *v*
  - The new nucleus contains *A* – 1 nucleons
  - The new nucleus has a speed of *u*
  - Each nucleon has a mass of *m*
- After the nucleus decays :
- Momentum of proton = mv
- Momentum of new nucleus = $A - 1m - u$
- Total momentum after decay = $mv - A - 1mu$
  - $0 = mv - A - 1mu$
  - $v = A - 1u$


Before
After
u
V
A nucleons
A-1 nucleons
The nucleus and the proton travel in opposite directions, so we can take V as positive and u as negative

Question 30 Hard [1 mark]

A particle X has speed *v* and collides with a stationary identical particle Y. The collision is perfectly elastic.

What are the speed and direction of motion of each of the two particles after the collision?

	X	Y
A	stationary	v to the right
B	$\frac{v}{2}$ to the right	$\frac{v}{2}$ to the right
C	$\frac{v}{2}$ to the left	$\frac{v}{2}$ to the right
D	*v* to the left	stationary

A

B

C

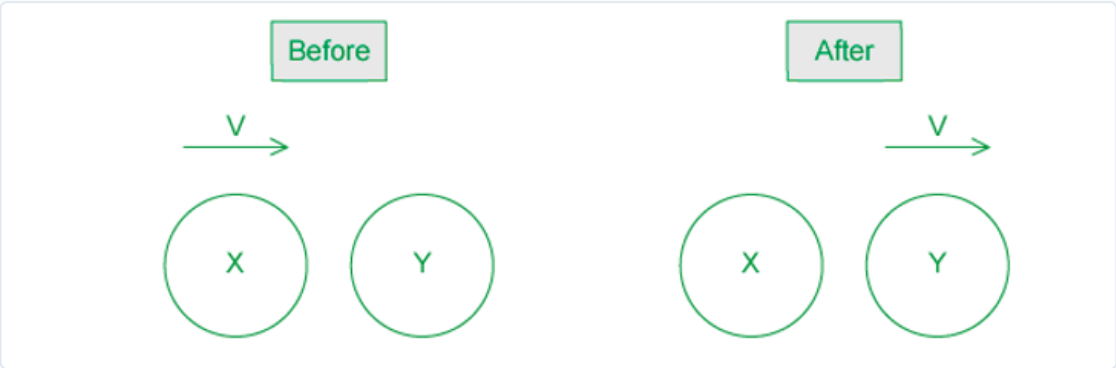
D

Mark Scheme & Explanation:

*The correct answer is **A** because:*

- In a perfectly elastic collision, both momentum and energy are conserved
- According to conservation of momentum:
  - Sum of momentum before collision = Sum of momentum after collision
- In an elastic collision, the total kinetic energy before the collision should be the same as the total kinetic energy after
- Consider option A :
- Before collision :
  - Momentum = $mv + 0 = mv$
  - Kinetic energy = $\frac{1}{2}mv^2 + 0 = \frac{1}{2}mv^2$
- After collision:
  - Momentum = $0 + mv = mv$
  - Kinetic energy = $0 + \frac{1}{2}mv^2 = \frac{1}{2}mv^2$

- Momentum and energy are both conserved so option A is correct



B is incorrect because

- Momentum : $m\frac{1}{2}v + m\frac{1}{2}v = mv$
- Kinetic energy : $\frac{1}{2}m\frac{1}{2}v^2 + \frac{1}{2}m\frac{1}{2}v^2 = \frac{1}{4}mv^2$

C is incorrect because

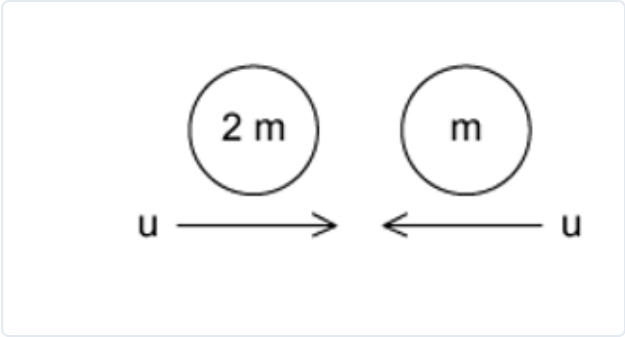
- Momentum : $m\frac{1}{2}v$ - $m\frac{1}{2}v = 0$
- Kinetic energy : $\frac{1}{2}m\frac{1}{2}v^2 + \frac{1}{2}m\frac{1}{2}v^2 = \frac{1}{4}mv^2$

D is incorrect because

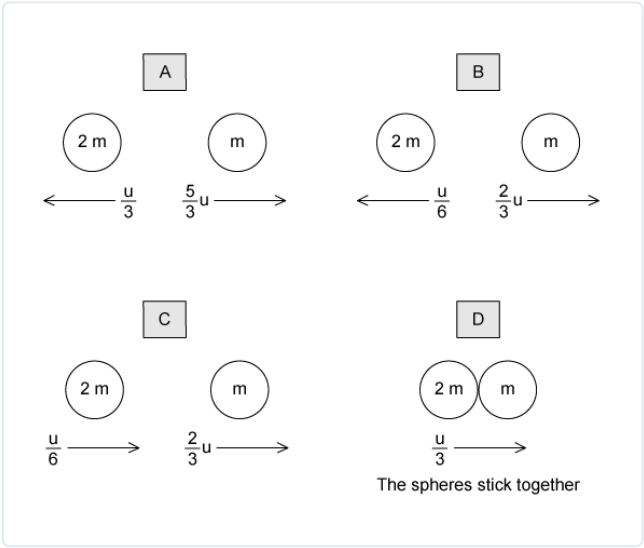
- Momentum : 0 - mv = - mv
- Kinetic energy : $0 + \frac{1}{2}mv^2 = \frac{1}{2}mv^2$

Question 31 Hard [1 mark]

The diagram shows two spherical masses approaching each other head-on at an equal speed *u*. One is of mass *m* and the other of mass 2 *m*.



Which diagram, showing the situation after the collision, is not consistent with the principle of conservation of momentum?



A

B

C

D

Mark Scheme & Explanation:

*The correct answer is **B** because:*

- According to conservation of momentum:
  - Sum of momentum before collision = Sum of momentum after collision
- Sum of momentum before collision = $2mu$ - mu = mu

- Sum of momentum after collision
  - $\frac{2m\text{-}u}{6} + \frac{2mu}{3} = \frac{2mu}{3} - \frac{2mu}{6} = \frac{2mu}{3} - \frac{mu}{3} = \frac{mu}{3}$
- The momentum before and after the collision is different so momentum has not been conserved in Option B

A is incorrect because momentum has been conserved

sum of momentum after collision : $\frac{5mu}{3} - \frac{2mu}{3} = \frac{3mu}{3} = mu$ ✓

C is incorrect because momentum has been conserved

sum of momentum after collision : $\frac{2mu}{3} + \frac{2mu}{6} = \frac{2mu}{3} + \frac{mu}{3} = \frac{3mu}{3} = mu$ ✓

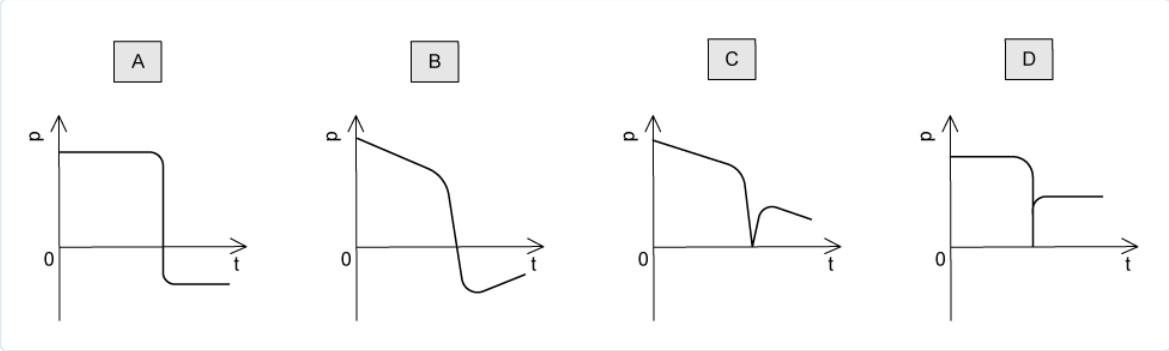
D is incorrect because momentum has been conserved

sum of momentum after collision : $\frac{2m + mu}{3} = \frac{3mu}{3} = mu$ ✓

Question 32 Hard [1 mark]

An ice-hockey puck slides along a horizontal, frictionless ice-rink surface. It collides inelastically with a wall at right angles to its path, and then rebounds along its original path.

Which graph shows the variation with time *t* of the momentum *p* of the puck?



A

B

C

D

Mark Scheme & Explanation:

*The correct answer is **A** because:*

- The fact that the ice-rink is frictionless is significant because it means the speed of the ice-hockey puck is not affected by friction and can be assumed to travel with constant speed before the collision
- This is represented on the graph by a straight horizontal line, since momentum is equal to $p = mv$
- After the collision, the puck will rebound along its original path and its momentum will be negative s it travels in the opposite direction
- The collision is inelastic, so the puck loses some energy and will rebound with a lower speed
- On the graph, at a later time the line will be straight and horizontal but below the x-axis and will have a lower value of momentum

B & C are incorrect because the speed is constant, so the momentum is constant and this should be represented by straight, horizontal lines on the graph

D is incorrect because momentum is a vector, so if the original direction is taken as positive, the direction after the rebound is negative and should be below the x-axis

Question 33 Hard [1 mark]

A body, initially at rest, explodes into two masses M_1 and M_2 that move apart with speeds v_1 and v_2 respectively.

What is the ratio $\frac{v_1}{v_2}$?

A $\frac{M_2}{M_1}$

B $\frac{M_1}{M_2}$

C $\sqrt{\frac{M_2}{M_1}}$

D $\sqrt{\frac{M_1}{M_2}}$

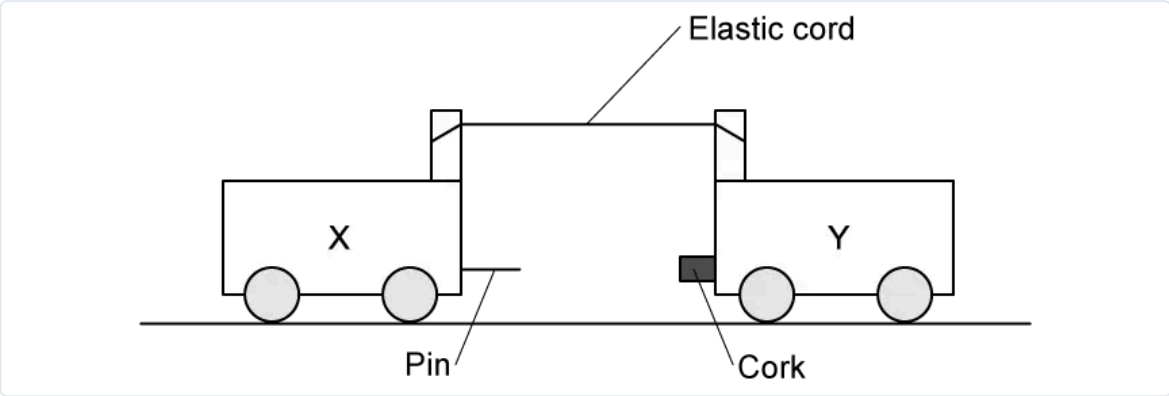
Mark Scheme & Explanation:

*The correct answer is **A** because:*

- The principle of conservation of momentum can be written as
  - $M_1 u_1 + M_2 u_2 = M_1 v_1 + M_2 v_2$
- The two masses are both initially at rest so
  - $u_1 = u_2 = 0$
  - $M_1 v_1 + M_2 v_2 = 0$
- The masses move away from each other in opposite directions, so let's define v_1 as the positive direction and v_2 as the negative direction
  - $M_1 v_1 - M_2 v_2 = 0$
  - $M_1 v_1 = M_2 v_2$
  - $M_2 = \frac{v_1}{v_2} M_1$
  - $\frac{v_1}{v_2} = \frac{M_2}{M_1}$

Question 34 Hard [1 mark]

Two trolleys, X and Y, are connected by an elastic cord, of negligible mass, as shown below. Trolley X is twice the mass of trolley Y.



The two trolleys are pulled apart so that the elastic cord is stretched. The trolleys are then released at the same time and move towards each other. The trolleys stick together on impact.

Just before the trolleys collide, the speed of trolley X is 20 cm s^{-1}.

What is the speed of the trolleys after they collide?

A 0 cm s^{-1}

B 10 cm s^{-1}

C 20 cm s^{-1}

D 30 cm s^{-1}

Mark Scheme & Explanation:

*The correct answer is **A***

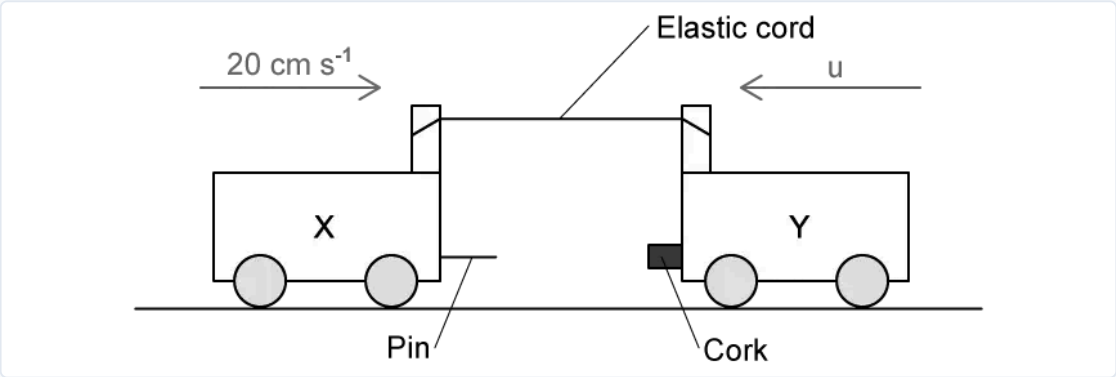
- If we consider the trolleys and the elastic cord as one system, there are no external forces acting
- Before the trolleys were released, they were at rest
  - Therefore, the total momentum of the system is **zero**
- Momentum is conserved in all collisions
  - Therefore, the total momentum of the system of trolleys and elastic cord after the collision will be **zero**
- As the trolleys stick together, the total momentum of the system after the collision is calculated using:

- (mass X + mass Y) × common velocity of both trolleys
- Hence, (mass X + mass Y) × common velocity of both trolleys = 0

- Therefore, the common velocity of both trolleys is **zero**

ALTERNATIVE METHOD

- The total momentum before the trolleys collide will be zero
- Before the impact, trolley X is travelling at 20 cm s^{-1}



- From the conservation of momentum:

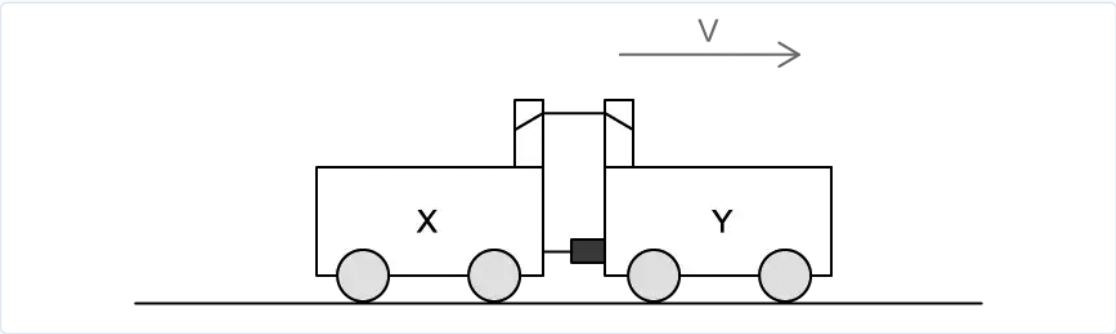
$$(mass \times velocity)_X + (mass \times velocity)_Y = 0$$

$$(2m \times 20) - (m \times u) = 0$$

$$40m = mu$$

$$u = 40 \text{ cm } s^{-1}$$

- Hence, trolley Y is travelling at 40 cm s^{-1} before the collision
- After the trolleys collide, they stick together
- Let the common velocity of the trolleys after the collision be v



- From conservation of momentum:

$$(mass \text{ X} + mass \text{ Y}) \times v = 0$$

$$(2m + m) \times v = 0$$

$$3mv = 0$$

- Hence, *v* must be **zero**